Physics 5C Discussion (Worksheet 2)

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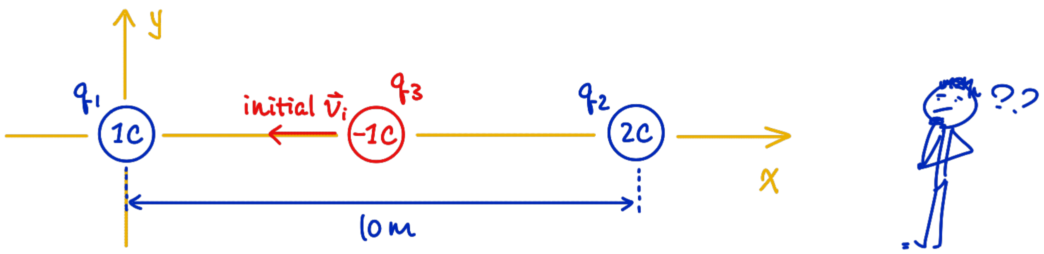
Click for the Zoom link and solutions to the worksheets.

Problem 1: There are two point charges q_1 and q_2 fixed in space. Now Jinyan throws a third point charge q_3 into the region, such that q_3 will have some initial velocity, as indicated in the figure.



$q_1 = 1\,\text{C}$ is fixed at $\vec{x}_1 = (0, 0)\,\text{m}$.
$q_2 = 2\,\text{C}$ is fixed at $\vec{x}_2 = (10, 0)\,\text{m}$.
$q_3 = -1\,\text{C}$ is initially between q_1 and q_2 on the x-axis with some initial velocity.
The Coulomb constant $k = 8.99 \cdot 10^9\,\text{Nm}^2/\text{C}^2$.

Part (a) Sketch the electric field produced by the source charges q_1 and q_2.

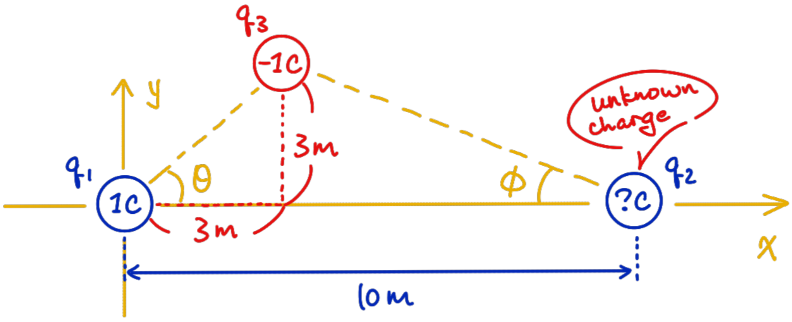
Part (b) At which point in space does the test charge q_3 experience a zero net force (neglect gravity and friction, only consider the electric force)?

Part (c) When q_3 is located at the point where it experiences zero net force, find the electric force experienced by q_2 in that configuration.

Part (d) Suppose q_3 is initially at $\vec{x}_3 = (5, 0)\,\text{m}$, not the point where it experiences zero net force, with initial velocity $\vec{v}_i = (-1, 0)\,\text{m/s}$. Qualitatively describe the motion of q_3 in such a configuration. How will it move under the electric forces from the source charges? Is there any way we can describe its motion quantitatively? [1]

[1] No need to do any calculation for this part, just describe in words.

Problem 2: [This problem is not too difficult, try to solve it when you get time.] Now we change the configuration a little bit, see the figure.

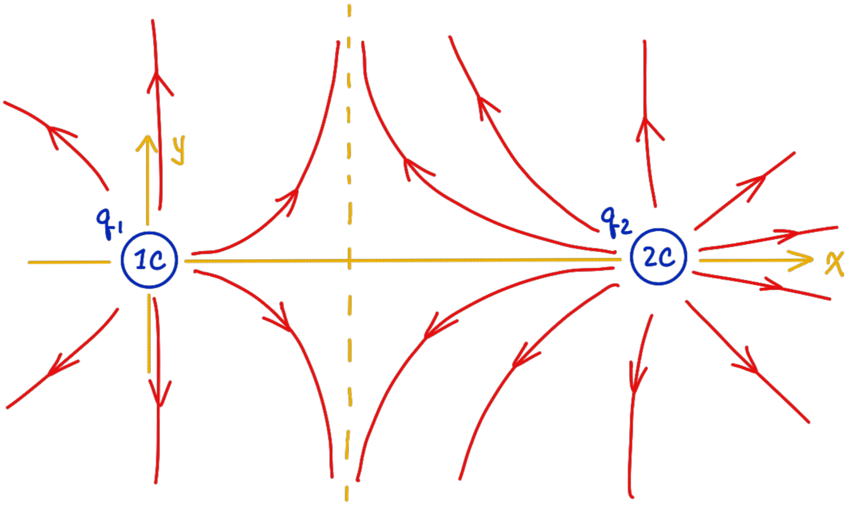


Now in this case, suppose we do not know the charge (or the sign) of q_2 fixed at $\vec{x}_2 = (10, 0)\,\mathrm{m}$, but we know that $q_1 = 1\,\mathrm{C}$ is fixed at $\vec{x}_1 = (0, 0)\,\mathrm{m}$ and $q_3 = -1\,\mathrm{C}$ placed at $\vec{x}_3 = (3, 3)\,\mathrm{m}$ has a tendency to move directly downwards (the electric force it experiences pulls it directly downwards). Find the electric field $\vec{E} = (E_x, E_y)$ at the position $\vec{x}_3 = (3, 3)\,\mathrm{m}$, due to the source charges q_1 and q_2. [2]

[2]Hint: q_3 wants to move directly downwards, that means the horizontal components of the forces acting on q_3 sum up to zero. There are only two electric forces acting on q_3, by the other two charges. Use some trigonometry to decompose the forces acting on q_3, then find the charge of q_2, and use that to find the electric field at $\vec{x}_3$ by superpositioning the electric fields due to q_1 and q_2.

Problem 1

Field lines come out from positive charges and terminate at negative charges. Also note that $q_2 = 2\,\mathrm{C}$ is "stronger" than $q_1 = 1\,\mathrm{C}$.



Here q_3 experiences two forces, one due to q_1 and the other one due to q_2. Since q_3 is negative, while q_1 and q_2 are positive, we deduce that both q_1 and q_2 are "pulling" q_3. In particular, q_1 is dragging q_3 towards q_1 on the left, and q_2 is dragging q_3 towards q_2 on the right. Therefore, the location where q_3 experiences zero net force is the place where the two forces due to q_1 and q_2 cancel out exactly. It is not hard to see that this location must be somewhere between q_1 and q_2 on the horizontal x-axis.

To find such a location, let us suppose q_3 experiences zero net force when it is located at $(x, 0)$ with $0 < x < 10$ (because we want q_3 to be between q_1 and q_2). Then its distance to q_1 is x, and its distance to q_2 is $10 - x$, so we can write the two forces

$$\vec{F}_{13} = -\frac{k|q_1 q_3|}{x^2}\hat{x}\,, \qquad \vec{F}_{23} = \frac{k|q_2 q_3|}{(10-x)^2}\hat{x}\,.$$

The first one is negative because q_1 is dragging q_3 to the left. Both forces have their y-components vanish because q_1 and q_2 want to drag q_3 directly towards them. The net force q_3 experiences is thus the sum

$$\vec{F}_{\text{net}} = \vec{F}_{13} + \vec{F}_{23} = kq_3\left(\frac{|q_2|}{(10-x)^2} - \frac{|q_1|}{x^2}\right)\hat{x}\,.$$

We want to find where $\vec{F}_{\text{net}} = (0, 0)$, thus it suffices to find where

$$\frac{|q_2|}{(10-x)^2} - \frac{|q_1|}{x^2} = 0\,.$$

Now we plug in the numbers,

$$\frac{2}{(10-x)^2} - \frac{1}{x^2} = 0\,.$$

There are many ways to solve for x, and here is one of them,

$$\begin{aligned}\frac{2}{(10-x)^2} &= \frac{1}{x^2}\,,\\ 2x^2 &= (10-x)^2\,,\\ 2x^2 &= 100-20x+x^2\,.\end{aligned}$$

We obtain a quadratic equation, using the quadratic formula we obtain

$$x = 4.14\,\mathrm{m}\,.$$

There is another solution $x = -24.14\,\mathrm{m}$, but that one is not physical.

Now we want to find the electric force acting on q_2, if we place q_3 at $\vec{x}_3 = (4.14,\ 0)\,\mathrm{m}$. In this case, since we want to find the force on q_2, the "other" charges that are pulling or pushing q_2 are q_1 and q_3. Therefore, q_2 is treated as a test charge, experiencing forces by the source charges q_1 and q_3. Here q_1 and q_2 have the same sign, so they repel each other, and thus q_2 experiences a force to the right (positive sign) due to q_1,

$$\vec{F}_{12} = \frac{k|q_1q_2|}{(\text{distance between } q_1 \text{ and } q_2)^2}\,\hat{x} = \left(\frac{8.99\cdot 10^9\cdot 1\cdot 2}{10^2}\,\mathrm{N}\right)\,\hat{x} = \left(1.80\cdot 10^8\,\mathrm{N}\right)\,\hat{x}\,.$$

On the other hand, q_3 and q_2 have opposite signs, thus q_3 is pulling q_2 to the left (negative sign for the force). Now we write

$$\vec{F}_{32} = -\frac{k|q_3q_2|}{(\text{distance between } q_3 \text{ and } q_2)^2}\,\hat{x} = \left(-\frac{8.99\cdot 10^9\cdot|-1|\cdot 2}{(10-4.14)^2}\,\mathrm{N}\right)\,\hat{x} = \left(-5.24\cdot 10^8\,\mathrm{N}\right)\,\hat{x}\,.$$

Adding these two forces together we get the net force experienced by q_2,

$$\vec{F}_{\text{net}} = \vec{F}_{12} + \vec{F}_{32} = \left(-3.44\cdot 10^8\,\mathrm{N}\right)\hat{x}\,.$$

After summing the effects due to both q_1 and q_3, we see that q_2 is getting pulled to the left (hence the negative sign in the net force). This is because q_2 is closer to q_3 which pulls q_2 to the left.

In the problem, it is said that q_3 is initially at $\vec{x}_3 = (5,0)$, to the right of the point where it experiences zero net force. Therefore, q_2 (on the right) would have a stronger effect on q_3, trying to pull q_3 to the right. However, q_3 is initially moving to the left with $\vec{v}_i = (-1,0)\,\mathrm{m/s}$, so q_3 will be moving to the left but slowing down. Depending on whether q_3 can pass through $x = 4.14\,\mathrm{m}$ (the point where the forces due q_1 and q_2 balance out), there will be two different end results:

1. q_3 is initially moving to the left, slowing down, able to pass through $x = 4.14\,\mathrm{m}$. In this case it enters the region where the force due to the left charge q_1 dominates, which pulls q_3 towards q_1, so q_3 will end up hitting q_1.

2. q_3 is initially moving to the left, slowing down, but unable to pass through $x = 4.14\,\mathrm{m}$. Then in this case, it will "turn around" before reaching $x = 4.14\,\mathrm{m}$, and remain in the region where the force due to the right charge q_2 dominates, which pulls q_3 towards q_2, so q_3 will end up hitting q_2.

Problem 2

In this configuration, we know that q_3 experiences a net force that has a zero x-component, because the net force is directly downwards. The force that q_1 exerts on q_3 has the form

$$\vec{F}_{13} = \left(-\frac{k|q_1 q_3|}{r^2} \cos(\theta)\,,\ -\frac{k|q_1 q_3|}{r^2} \sin(\theta) \right)\,,$$

where $r = \sqrt{3^2+3^2}\,\mathrm{m} = 3\sqrt{2}\,\mathrm{m}$ indicates the distance between charge q_1 and charge q_3. The sin and cos are there to extract the x- and y-components of $\vec{F}_{13}$. We are able to do this (extract the components) because the magnitude of $\vec{F}_{13}$ is $|kq_1q_3|/r^2$ based on Coulomb's law. Similarly, we write down

$$\vec{F}_{23} = \left(\frac{k|q_2 q_3|}{R^2} \cos(\phi)\,,\ -\frac{k|q_2 q_3|}{R^2} \sin(\phi) \right)$$

where $R = \sqrt{(10-3)^2+3^2} = \sqrt{58}\,\mathrm{m}$, and ϕ is the angle indicated on the original figure. The additional negative sign is added in the y-component to ensure that $\vec{F}_{23}$ is dragging q_3 towards q_2 (so we want the force to be pointing right-down, positive in x-component and negative in y-component).

Using the trigonometry here, we see that

$$\sin(\theta) = \frac{3}{\sqrt{3^2+3^2}} = \frac{1}{\sqrt{2}}\,, \qquad \cos(\theta) = \frac{3}{\sqrt{3^2+3^2}} = \frac{1}{\sqrt{2}}\,.$$

$$\sin(\phi) = \frac{3}{\sqrt{(10-3)^2+3^2}} = \frac{3}{\sqrt{58}}\,, \qquad \cos(\phi) = \frac{(10-3)}{\sqrt{(10-3)^2+3^2}} = \frac{7}{\sqrt{58}}\,.$$

So we conclude

$$\vec{F}_{13} = \left(-\frac{k|q_1 q_3|}{18}\frac{1}{\sqrt{2}}\,,\ -\frac{k|q_1 q_3|}{18}\frac{1}{\sqrt{2}} \right)\,, \qquad \vec{F}_{23} = \left(\frac{k|q_2 q_3|}{58}\frac{7}{\sqrt{58}}\,,\ -\frac{k|q_2 q_3|}{58}\frac{3}{\sqrt{58}}\,. \right)$$

The statement in the problem is that the sum of these two forces is directly downwards, that is the x-components should cancel each other,

$$-\frac{k|q_1 q_3|}{18}\frac{1}{\sqrt{2}} + \frac{k|q_2 q_3|}{58}\frac{7}{\sqrt{58}} = k|q_3| \left(-\frac{|q_1|}{18\sqrt{2}} + \frac{7|q_2|}{58\sqrt{58}} \right) = 0\,,$$

from which we can find q_2 by plugging in the value of q_1,

$$-\frac{1}{18\sqrt{2}} + \frac{7|q_2|}{58\sqrt{58}} = 0\,,$$

thus $q_2 = 2.48\,\mathrm{C}$ (the sign of q_2 has to be positive, otherwise the x-components cannot cancel each other out based on the geometry, as both forces will be pushing q_3 to the left if q_2 was negative). Now the electric field at $\vec{x}_3 = (3,3)\,\mathrm{m}$ is in fact the electric force experienced by q_3 divided by q_3 (note that q_3 is negative), that is,

$$\vec{E}(\vec{x}_3) = \frac{\vec{F}_{13}+\vec{F}_{23}}{q_3} = \left(\frac{k|q_1|}{18\sqrt{2}} - \frac{k|q_2|}{58}\frac{7}{\sqrt{58}}\,,\ \frac{k|q_1|}{18\sqrt{2}} + \frac{k|q_2|}{58}\frac{3}{\sqrt{58}} \right)\ \mathrm{N/C}\,.$$

Plug in $q_1 = 1\,\mathrm{C}$ and $q_2 = 2.48\,\mathrm{C}$ we find

$$\vec{E}(\vec{x}_3) = \left(0,\ 5.04 \cdot 10^8\right)\ \mathrm{N/C}\,.$$

A squirrel at the University of Michigan ... $\mathbb{GO}$ $\mathbb{BLUE}$!